

Sentiment Analysis of Amazon Product Reviews: Comparing Traditional and Modern Text Mining Approaches

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Abstract

This work addresses binary sentiment classification on Amazon food reviews, comparing traditional text mining representations against modern embedding-based approaches. We build a complete pipeline covering exploratory linguistic analysis (POS-tagging, Named Entity Recognition, n-gram extraction, WordNet), text preprocessing (tokenisation, stopword removal, lemmatisation, stemming), multiple feature representations (Term Frequency, TF-IDF with unigrams through trigrams), and sentence-level BERT embeddings. Three classifiers—Naive Bayes, Logistic Regression, and Support Vector Machines—are evaluated across all representations using 5-fold stratified cross-validation on a balanced dataset of 10,000 reviews. Our best model, TF-IDF with trigrams and a linear SVM, achieves a macro F1-score of 0.871 on the held-out test set, substantially outperforming the 0.333 majority-class baseline. Notably, traditional sparse representations consistently outperform BERT sentence embeddings (F1=0.798), suggesting that for well-structured review text with sufficient training data, bag-of-words models remain highly competitive. An ablation study further reveals that aggressive text preprocessing slightly degrades performance compared to simple lower-casing, indicating that sentiment-bearing function words and morphological variants carry useful discriminative signal.

1 Introduction

Sentiment analysis is one of the most widely applied tasks in text mining, aiming to determine whether a piece of text expresses a positive or negative opinion. It has direct applications across many domains: companies use it to monitor brand perception, product managers rely on it to identify recurring complaints, and economists leverage it to gauge consumer confidence from textual data.

The volume of user-generated text has grown exponentially with the rise of e-commerce platforms. Amazon alone hosts hundreds of millions of product

reviews, making manual analysis impossible. Automated sentiment classification provides a scalable alternative, enabling organisations to extract actionable insights from unstructured text data.

In this project, we tackle binary sentiment classification on Amazon food reviews, framing the problem as a systematic comparison of text representation strategies. The central question is: *does the choice of text representation—from simple bag-of-words to pre-trained BERT embeddings—significantly impact classification performance?* This question is particularly relevant given the growing adoption of neural language models, which are often assumed to outperform traditional methods without rigorous comparison.

Our contribution is a complete, reproducible text mining pipeline that covers the full spectrum of techniques: linguistic analysis at the morphological, syntactic, and semantic levels; statistical representations including TF, TF-IDF, and n-grams; and modern dense representations via sentence embeddings. We evaluate these on a balanced dataset of 10,000 reviews using cross-validation and macro F1-score, producing results that are statistically robust and practically meaningful.

The remainder of this paper is organised as follows. Section 2 provides background on sentiment analysis and text representation. Section 3 describes our methodology. Section 4 details the experimental setup. Section 5 presents and analyses the results. Section 6 concludes with a discussion of findings and future directions.

2 Background

Sentiment analysis has been studied extensively since the early 2000s. Pang et al. (2002) pioneered the use of machine learning for sentiment classification, showing that Naive Bayes and SVM classifiers with bag-of-words features could effectively classify movie reviews. This established a paradigm that remains influential: represent text as a vector of word-level features, then apply a supervised classifier.

The **bag-of-words** (BOW) model treats a document as an unordered collection of words, discarding

grammar and word order (Aggarwal and Zhai, 2012). Despite its simplicity, BOW remains a strong baseline. A key limitation is that it assigns equal importance to all words, including common function words that carry little discriminative information.

TF-IDF weighting (Salton and Buckley, 1988) addresses this by combining term frequency with inverse document frequency. Words that appear frequently in a specific document but rarely in others receive higher weights, effectively highlighting content-bearing vocabulary while suppressing ubiquitous terms.

N-gram features extend the unigram model by considering sequences of consecutive words. This partially recovers word order information: the bigram “not good” carries negative sentiment that neither “not” nor “good” expresses individually. However, increasing the n-gram order exponentially expands the feature space, requiring dimensionality limits.

More recently, **word embeddings** such as Word2Vec (Mikolov et al., 2013) and contextual embeddings from BERT (Devlin et al., 2019) have transformed NLP by representing words as dense vectors. Sentence-level embeddings (Reimers and Gurevych, 2019) compress entire sentences into fixed-size vectors, enabling direct use with traditional classifiers.

A persistent concern is **class imbalance**. Real-world review datasets are heavily skewed toward positive reviews (He and Garcia, 2009), which can lead to models that simply predict the majority class. Addressing this through balanced sampling and appropriate metrics is essential.

3 Methodology

Our methodology follows a structured pipeline consisting of five stages: exploratory linguistic analysis, text preprocessing, feature representation, classification, and error analysis.

3.1 Exploratory Linguistic Analysis

Before building any classifier, we perform linguistic analysis using SpaCy and NLTK:

POS-tagging: Using SpaCy’s `en_core_web_sm` model, we tag each word with its part of speech and compare distributions across positive and negative reviews.

Named Entity Recognition (NER): We extract named entities (organisations, dates, cardinal numbers) from both classes.

N-gram analysis: We extract the 15 most frequent bigrams and trigrams for each sentiment class using scikit-learn’s `CountVectorizer`.

WordNet: Using NLTK’s WordNet interface, we explore synonym and antonym relationships for key sentiment words (e.g., *good*, *bad*, *excellent*, *terrible*, *love*, *hate*), illustrating the variability problem in natural language.

3.2 Text Preprocessing

Each review undergoes the following pipeline using SpaCy:

1. **Lowercasing:** All text is converted to lowercase.
2. **Punctuation removal:** Punctuation marks are discarded.
3. **Stopword removal:** Common English words are filtered out.
4. **Lemmatisation:** Words are reduced to their base form (e.g., “running” → “run”).

We also conduct an **ablation study** comparing four preprocessing variants—lowercase only, stopword removal, stemming (Porter Stemmer), and full lemmatisation—to determine which steps actually improve classification.

3.3 Feature Representation

We construct five sparse representations: **TF Unigrams**, **TF Uni+Bigrams**, **TF-IDF Unigrams**, **TF-IDF Uni+Bigrams**, and **TF-IDF Uni+Bi+Trigrams** (capped at 80,000 features).

Additionally, we generate **sentence embeddings** using the pre-trained `all-MiniLM-L6-v2` Sentence-Transformer model (Reimers and Gurevych, 2019), producing 384-dimensional dense vectors.

3.4 Classification

Three classifiers are evaluated: **Multinomial Naive Bayes**, **Logistic Regression** (L2 regularisation), and **Linear SVM**. All use scikit-learn with default hyperparameters and fixed random seeds for reproducibility.

4 Experimentation

4.1 Dataset

We use the Amazon Fine Food Reviews dataset (McAuley and Leskovec, 2013), containing 568,454 reviews with star ratings (1–5). After removing neutral reviews (score = 3) and missing values, we obtain 525,814 reviews: 443,777 positive and 82,037 negative (5.4:1 ratio). We create a **balanced sample** of 5,000 reviews per class (10,000 total) via undersampling, ensuring a trivial baseline achieves only 50% accuracy and 0.333 macro F1.

4.2 Evaluation Protocol

The balanced dataset is split 80/20 into training (8,000) and test (2,000) using stratified sampling. We perform **5-fold stratified cross-validation** on the training set. The primary metric is **macro F1-score**, which treats both classes equally. A majority-class baseline is included for reference.

5 Results and Analysis

5.1 Exploratory Findings

The review length analysis (Figure 1) shows similar word count distributions for both classes, indicating that length alone does not discriminate sentiment.

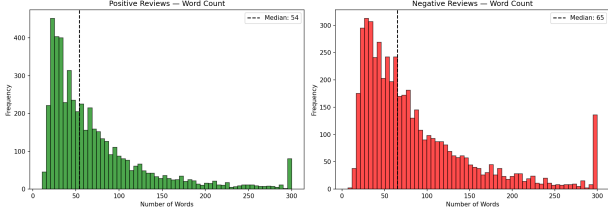


Figure 1: Word count distribution for positive and negative reviews.

The word cloud analysis (Figure 2) reveals clear lexical differences. Positive reviews feature *love*, *great*, *delicious*, while negative reviews show *disappointed*, *waste*, *bland*. Words like *taste* appear in both clouds with different connotations, illustrating the ambiguity problem.

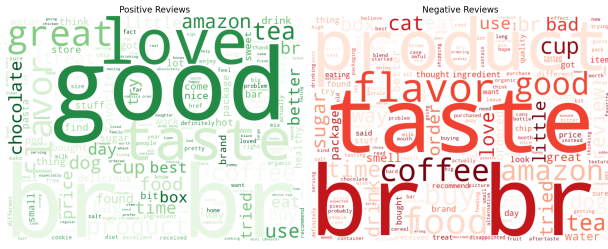


Figure 2: Word clouds for positive (left) and negative (right) reviews.

The n-gram analysis (Figure 3) identifies sentiment-bearing expressions: “great taste” and “highly recommend” (positive) versus “waste money” and “taste like” (negative). These motivate the inclusion of n-gram features.

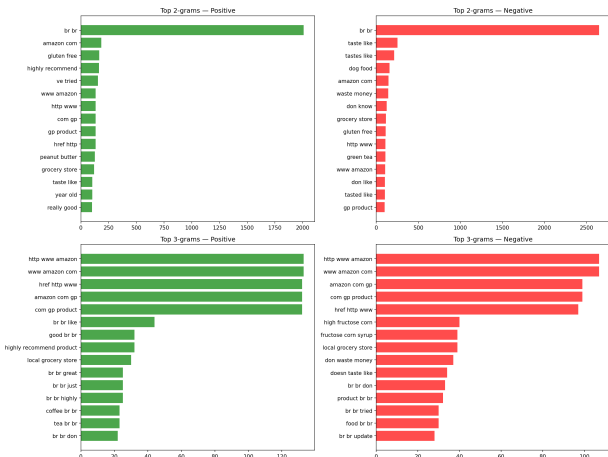


Figure 3: Top bigrams (top) and trigrams (bottom) for positive (left) and negative (right) reviews.

The POS-tag analysis (Figure 4) shows remarkably similar syntactic structures across classes, suggesting that sentiment is expressed through lexical choice rather than syntactic structure, supporting the bag-of-words approach.

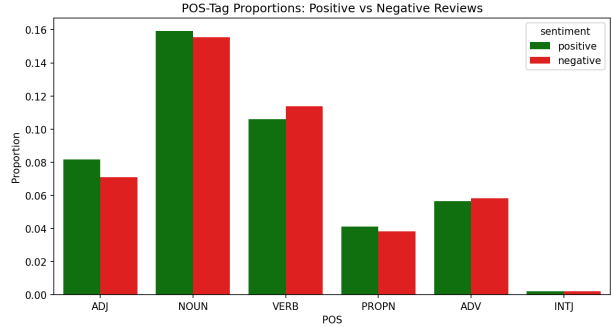


Figure 4: POS-tag proportions across sentiment classes.

The NER analysis (Figure 5) reveals that cardinal numbers and organisations are the most common entity types, reflecting the food review domain.

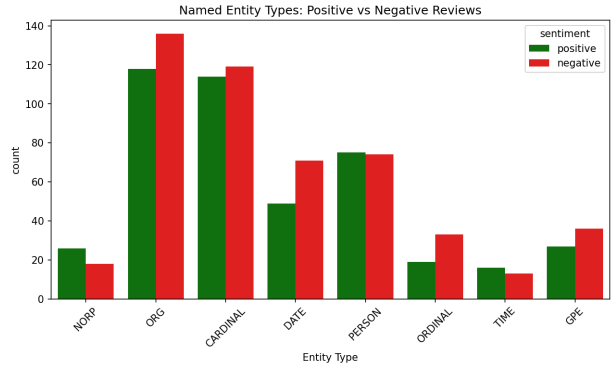


Figure 5: Named entity type distribution by sentiment class.

5.2 Classification Results

Table 1 presents the macro F1-scores for all combinations. All models substantially outperform the baseline (F1=0.333). Figure 6 provides a visual comparison.

Table 1: Test macro F1-scores for all combinations. Best in bold.

Representation	NB	LR	SVM
TF Unigrams	.824	.848	.828
TF Uni+Bigrams	.843	.864	.856
TF-IDF Unigrams	.839	.856	.855
TF-IDF Uni+Bigrams	.863	.848	.867
TF-IDF Uni+Bi+Tri	.862	.850	.871
Sent. Emb. (BERT)	—	.798	—

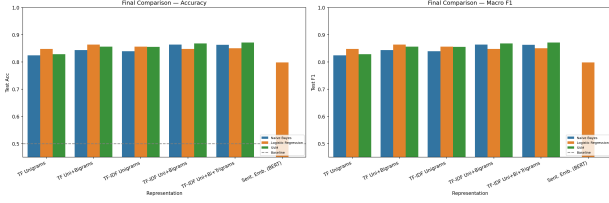


Figure 6: Test accuracy and macro F1 across all approaches. Dashed line: majority-class baseline.

The best model is **TF-IDF Uni+Bi+Trigrams with Linear SVM** (F1=0.871, CV F1=0.858±0.010). Per-class metrics (Table 2) confirm balanced performance. The confusion matrix (Figure 7) shows errors are equally distributed.

Table 2: Per-class metrics for the best model (TF-IDF trigrams + SVM).

Class	Prec.	Recall	F1
Negative	0.86	0.88	0.87
Positive	0.88	0.86	0.87
Macro avg	0.87	0.87	0.87

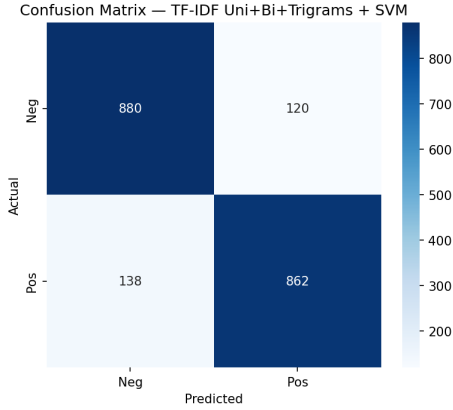


Figure 7: Confusion matrix for the best model.

5.3 Analysis of Representation Strategies

Three clear patterns emerge:

TF-IDF outperforms raw TF. Across all classifiers, TF-IDF yields higher F1 than TF. For instance, TF-IDF unigrams with LR (0.856) outperforms TF unigrams with the same classifier (0.848). The IDF component successfully down-weights common words and emphasises discriminative terms.

N-gram features improve performance. Adding bigrams and trigrams consistently helps. The improvement is most pronounced for SVM: TF-IDF unigrams (0.855) versus TF-IDF trigrams (0.871). N-gram features capture negation (“not good”), intensifiers (“very disappointed”), and multi-word expressions (“waste of money”).

Traditional sparse methods outperform BERT embeddings. Sentence embeddings achieve F1=0.798, substantially below all sparse representations. The pre-trained model was not fine-tuned on this domain, and compressing each review into 384 dimensions discards lexical detail that 80,000-feature sparse vectors preserve.

5.4 Most Discriminative Features

Figure 8 shows the top 15 features per class. Positive indicators include *love*, *great*, *excellent*; negative indicators include *disappointed*, *waste*, *bland*. N-gram features like “highly recommend” and “waste money” appear among top discriminators, confirming their value.

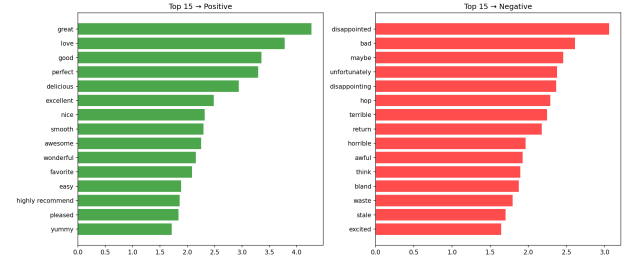


Figure 8: Top 15 most discriminative features for positive (left) and negative (right) sentiment.

5.5 Impact of Preprocessing

The ablation study (Figure 9) yields a counterintuitive result: **simple lowercasing alone (F1=0.851) outperforms all other preprocessing strategies**—stopword removal (0.813), stemming (0.813), and lemmatisation (0.809).

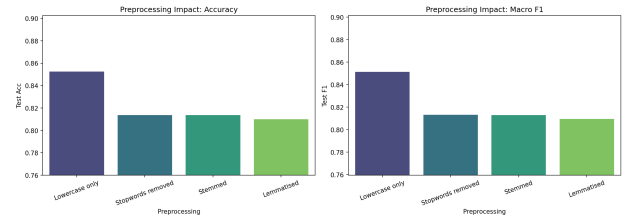


Figure 9: Impact of preprocessing strategies. Lowercase-only achieves the best performance.

This has a clear linguistic explanation. Stopword removal eliminates words critical for sentiment: negations (“not”, “never”), intensifiers (“very”, “really”, “too”), and contrastive conjunctions (“but”, “however”). Removing these destroys sentiment signals—“not good” becomes just “good”. Similarly, stemming and lemmatisation collapse morphological variants that carry different implications: “disappointing”, “disappointed”, and “disappoints” all reduce to the same root, losing temporal and aspectual nuances that help distinguish factual statements from opinions.

5.6 Error Analysis

Of 2,000 test reviews, 258 (12.9%) are misclassified. False positives (negative reviews predicted as positive) frequently contain mixed sentiment: “*I think it’s wonderful that this is 100% dairy free, but it has such a fake flavor.*” The bag-of-words model counts positive words alongside negative ones, and the positive vocabulary dominates.

False negatives (positive reviews predicted as negative) often use hedging: “*I really like the taste, just a touch of salt, but they are more chewy than I would like.*” Words like “but” and “chewy” trigger negative associations despite overall positive sentiment.

Shallow parsing reveals that noun phrases in misclassified reviews encode sentiment at the phrase level (“such a fake flavor of chocolate”), which unigram features cannot capture but n-grams partially address.

6 Conclusions

This project demonstrates a complete text mining pipeline for sentiment analysis on 10,000 balanced Amazon food reviews with robust cross-validation and appropriate metrics.

Our first finding is that **TF-IDF with n-gram features and SVM achieves the best performance** ($F1=0.871$), with balanced precision and recall. Both TF-IDF weighting and n-gram features contribute meaningfully.

Our second finding is that **pre-trained BERT sentence embeddings ($F1=0.798$) underperform traditional sparse representations**, challenging the assumption that neural embeddings are universally superior. Frozen, general-purpose embeddings lose domain-specific lexical signals that sparse models preserve.

Our third finding is that **text preprocessing can hurt sentiment classification**. Stopword removal eliminates sentiment-critical function words, and stemming/lemmatisation collapses useful morphological variants. For sentiment analysis, minimal preprocessing is preferable.

Future work should explore fine-tuning BERT directly on the classification task, testing on other domains (electronics, movies), investigating the effect of training set size on the sparse-vs-dense comparison, and incorporating techniques such as SMOTE for handling class imbalance.

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